

Department Overview Department Name: Department of Telecommunication Engineering Faculty: Faculty of Information & Communication Technology (FICT) Accreditation: The program is officially accredited by the Pakistan Engineering Council (PEC). Core Technological Focus: Cutting-edge research and application in emerging fields including Blockchain, Artificial Intelligence (AI), AI-enabled telecommunication networks & systems, and 5G & beyond networks. Industrial Specialization Disciplines: RF & Microwave, Satellite Communication, Optical Fiber Communication, Wireless & Mobile Communication, and Analog & Digital Communication. History & Impact: Began as an undergraduate program in 2005, graduating its first batch in 2009. It equips approximately 50 students with telecommunication systems expertise each year. The department expanded to offer an MS degree program in 2013. Program Educational Objectives (PEOs) The program is structured to cultivate technical experts who will:

1. Investigate engineering problems using modern techniques and propose effective solutions.
2. Manage projects and become effective members of engineering teams through interpersonal skills.
3. Demonstrate high moral and ethical values, a life-long learning attitude, and fulfill societal responsibilities.

Bachelor of Science (BS) Program

Admission Requirements F.Sc / Equivalent: F.Sc (Pre-Engineering) or ICS with Mathematics and Physics from a recognized board, securing at least 60% marks. Diploma of Associate Engineer (DAE): Completed in a relevant field with a minimum of 60% marks.

Degree Requirements Total Credit Hours: 130-145 Total Courses: 42 - 44 Minimum Passing Standing: CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory + Lab).

1st Semester • HUM-163 | Functional English | Prerequisite: None | Credit Hours: 3+0 • MATHP-118 | Calculus & Analytical Geometry | Prerequisite: None | Credit Hours: 3+0 • HUM-101 | Islamic Studies / Ethical Behavior | Prerequisite: None | Credit Hours: 2+0 • PHY-205 | Applied Physics | Prerequisite: None | Credit Hours: 3+1 • CE-162L | Computer Aided Engineering Design | Prerequisite: None | Credit Hours: 0+1 • CS-114 | Programming Fundamentals | Prerequisite: None | Credit Hours: 2+1 • EE-103L | Electronics Workshop | Prerequisite: None | Credit Hours: 0+1

2nd Semester • MATHA-214 | Differential Equations | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0 • HUM-268 | Communication Skills | Prerequisite: None | Credit Hours: 3+0 • EE-102 | Digital Logic Design | Prerequisite: None | Credit Hours: 3+1 • EE-203 | Circuit Analysis | Prerequisite: None | Credit Hours: 3+1 • CS-212 | Object Oriented Programming | Prerequisite: Programming Fundamentals | Credit Hours: 3+1

3rd Semester • MATHA-213 | Complex Variables & Transforms | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0 • HUM-102 | Pakistan Studies | Prerequisite: None | Credit Hours: 2+0 • ELE-201 | Electrical Network Analysis | Prerequisite: Circuit Analysis | Credit Hours: 2+1 • MATHP-111 | Linear Algebra | Prerequisite: None | Credit Hours: 3+0 • CS-214 | Data Structures and Algorithms (IDEE-I) | Prerequisite: None | Credit Hours: 2+1 • EE-208 | Electronic

Devices and Circuits | Prerequisite: Applied Physics | Credit Hours: 3+1

4th Semester • TE-408 | Computer Communications & Networks | Prerequisite: None | Credit Hours: 3+1 • CE-324 | Microprocessors and Microcontrollers | Prerequisite: Digital Logic Design | Credit Hours: 3+1 • ELE-202 | Probability Methods in Engineering | Prerequisite: None | Credit Hours: 3+0 • EE-261 | Signals & Systems | Prerequisite: Complex Variables and Transforms | Credit Hours: 3+0 • EE-105 | Electromagnetic Theory | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3+0

5th Semester • TE-433 | Wave Propagation & Antennas | Prerequisite: Electromagnetic Theory | Credit Hours: 3+1 • TE-314 | Communication Systems | Prerequisite: None | Credit Hours: 3+1 • HUM-264 | Technical Report Writing | Prerequisite: None | Credit Hours: 3+0 • TE-412 | Digital Signal Processing | Prerequisite: Signals and Systems | Credit Hours: 3+1 • EE-241 | Control Systems | Prerequisite: Linear Algebra | Credit Hours: 2+1

6th Semester • TE-442 | Optical Fiber Communications | Prerequisite: None | Credit Hours: 3+1 • TBA | IDEE-II | Prerequisite: None | Credit Hours: 2+1 • TE-313 | Digital Communications | Prerequisite: Probability Methods in Engineering | Credit Hours: 3+1 • MGMT-202 | Entrepreneurship & Leadership | Prerequisite: None | Credit Hours: 3+0 • TE-415 | Transmission and Switching Systems | Prerequisite: None | Credit Hours: 3+1

7th Semester • TE-432 | Microwave Engineering | Prerequisite: Wave Propagation & Antennas | Credit Hours: 3+1 • TE-425 | Emerging Wireless Technologies & RF Planning | Prerequisite: None | Credit Hours: 3+0 • TE-422 | MBC I (Telecom Traffic Engineering) | Prerequisite: Probability Methods in Engineering | Credit Hours: 3+0 • ECON-105 | Engineering Economics | Prerequisite: None | Credit Hours: 3+0 • TE-452L | Final Year Project-I | Prerequisite: None | Credit Hours: 0+3

8th Semester • MGMT-304 | Organizational Behavior or Sociology | Prerequisite: None | Credit Hours: 3+0 • TE-423 | MBC II (Telecom Policies & Regulation) | Prerequisite: None | Credit Hours: 3+0 • TE-441 / TE-409 | MBC III (Satellite Comm. or Network Security) | Prerequisite: None | Credit Hours: 3+0 • HUM-302 | Professional Practices | Prerequisite: None | Credit Hours: 3+0 • TE-453L | Final Year Project-II | Prerequisite: None | Credit Hours: 0+3